

One Function, Three Spellings

Where the types sit, how each line is read, and why one of them will not compile

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The function

All three lines below define the same thing: an async function that takes a number `x` and resolves to `x * 2`. Its type, in every case, is a function from a number to a promise of a number, written `(x: number) => Promise<number>`.

```
// Form 1: types on the function, variable inferred
const f = async (x: number): Promise<number> => { return x * 2 };

// Form 2: types on the variable, function bare
const f: (x: number) => Promise<number> = async (x) => { return x * 2 };

// Form 3: function declaration (no variable at all)
async function f(x: number): Promise<number> { return x * 2 }
```

Vocabulary used below

Word	Means	In these examples
the arrow / the arrow function	a function written with <code>=></code>	<code>async (x) => { ... }</code>
the async function	a function marked <code>async</code> ; always returns a Promise	the same thing, named by its keyword
the value / the right-hand side	what comes after <code>=</code>	the same thing, named by its position
the variable	the name being declared	<code>f</code>
the variable's type	what comes between <code>:</code> and <code>=</code> on the left	<code>(x: number) => Promise<number></code>
<code>Promise<number></code>	a placeholder that will later hold a number	the return type of any async function returning a number

Form 1: types on the function

```
const f = async (x: number): Promise<number> => { return x * 2 };
      ^      ^^^^^^ ^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^
      2      async parameter return type      => body
              keyword with type
```

Token by token

Piece	Say	Slot
<code>const f</code>	"constant <code>f</code> "	1, 2
<code>=</code>	"is set to"	border
<code>async</code>	"an async function"	4
<code>(x: number)</code>	"taking a number <code>x</code> "	4
<code>: Promise<number></code>	"that returns a promise of a number"	4
<code>=></code>	"whose body is"	4
<code>{ return x * 2 }</code>	"return <code>x</code> times 2"	4

Read aloud, whole:

"Constant `f` is set to an async function taking a number `x`, returning a promise of a number, whose body returns `x` times 2."

Where the types are:

both inside the function. The `:` after `x` types the parameter. The `:` after `)` types the return. The variable `f` has no annotation (no `:` between `f` and `=`); TypeScript infers `f`'s type from the arrow: `(x: number) => Promise<number>`.

What is optional here:

the `: Promise<number>`. Drop it and TS still infers `Promise<number>` because the function is `async` and returns a number. The `x: number` is not optional in strict mode: with nothing on the left to supply it, a bare `(x)` would be an implicit `any`.

Form 2: types on the variable

```
const f: (x: number) => Promise<number> = async (x) => { return x * 2 };
      ^               ^               ^               ^               ^
      2 3: the TYPE of f               = 4: the VALUE   => body
      (a function type)               async arrow, bare x
```

Token by token

Piece	Say	Slot
const f	"constant f"	1, 2
: (x: number) => Promise<number>	"of type: function from a number x to a promise of a number"	3
=	"is set to"	border
async	"an async function"	4
(x)	"taking x" (its type comes from the left)	4
=>	"whose body is"	4
{ return x * 2 }	"return x times 2"	4

Read aloud, whole:

"Constant f, of type function-from-number-to-promise-of-number, is set to an async function taking x whose body returns x times 2."

Where the types are:

everything between : and = is the type of f. The arrow on the right has no annotations at all. x is still a number, because TS reads the expected type on the left and pushes it into the bare parameter. This is called

contextual typing

, and it is the same mechanism that types x in arr.map((x) => x * 2): the expected callback type comes from map.

Two arrows, two jobs

```
const f: (x: number) => Promise<number> = async (x) => { return x * 2 };
//               ^^ "to"                               ^^ "whose body is"
//               | TYPE (after the colon) |             | VALUE (after the equals) |
```

Arrow	Side of =	Job	Say
(x: number) => Promise<number>	left, after :	describes a shape; runs nothing	"number to promise of number"
async (x) => { ... }	right, after =	a real function	"taking x, whose body is ..."

The = is the border. Left of it is what f IS; right of it is what f HOLDS.

Form 3: function declaration

```
async function f(x: number): Promise<number> { return x * 2 }
^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^
async keyword  name param  return type      body
```

Read aloud:

"Async function f, taking a number x, returning a promise of a number, whose body returns x times 2."

No const, no =, no arrow. The name f is part of the function syntax. The types sit in the same two places as Form 1 (after x, after). This is the most common form for a standalone, named function in the code you will review.

The broken one

```
const f: (x: number) => async (x) => { return x * 2 };    // x does not compile
```

Why:

there is no =. After the : TypeScript is reading a

type

and keeps reading until it finds `=`. It sees `(x: number) =>`, the start of a function type, and now expects a *return type*. It gets `async (x) => { ... }`, which is a value, not a type, so it stops with a syntax error. Slot 3 never ended; slot 4 never started.

```
const f: (x: number) => async (x) => { return x * 2 };
      2 3 ..... ?           no border, so no slot 4
```

The fix

is to insert the border and finish the type:

```
const f: (x: number) => Promise<number> = async (x) => { return x * 2 };
//                               ^
//                               finish the type the border
```

Side by side

	Form 1	Form 2	Form 3
Written	<code>const f = async (x: number): Promise<number> => { ... }</code>	<code>const f: (x: number) => Promise<number> = async (x) => { ... }</code>	<code>async function f(x: number): Promise<number> { ... }</code>
Where <code>x: number</code> sits	on the arrow	on <code>f</code> 's type	on the declaration
Where <code>Promise<number></code> sits	after the arrow's <code>)</code>	on <code>f</code> 's type, after <code>=></code>	after the declaration's <code>)</code>
Arrow parameters	typed	bare (contextually typed)	n/a
Has a <code>=</code>	yes	yes	no
Number of <code>=></code>	one (value)	two (one type, one value)	none
Type of <code>f</code>	inferred from the arrow	declared; the arrow must match it	from the declaration
<code>this</code> inside	the surrounding <code>this</code>	the surrounding <code>this</code>	depends on how it is called
Hoisted (usable before the line)	no	no	yes
Meaning	identical	identical	identical

Reading rules

1. **Find the `=`.** Left of it (after `:`) is the type and runs nothing. Right of it is the value. No `=`: either a **function** declaration (Form 3) or a callback passed as an argument, or a syntax error.
2. **Two `=>` on one line** means one is in the type and one is in the value; the `=` is between them. Say “to” for the type arrow and “whose body is” for the value arrow.
3. **A bare `(x)` is not `any`.** Ask where its type comes from: the variable’s annotation on the left (Form 2), or the method it is passed to (`map`, `then`, `app.get`). Only a bare parameter with nothing on either side is an implicit `any`.
4. **A missing `: Promise<number>` is not a bug.** It is inferred. To work it out yourself: find every `return`; if the function is `async`, wrap the result in `Promise<...>`.
5. **`async` means the return type always starts with `Promise<`**, whether written or inferred. Every caller must `await` it; a caller that uses the result directly is holding the promise, not the value.

Where each form shows up in real code

You will see	Form	Note
<code>export async function getUser(id: string): Promise<User></code>	3	most exported functions
<code>const handler = async (req: Request, res: Response) => { ... }</code>	1	handlers assigned to a const
<code>const handler: RequestHandler = async (req, res) => { ... }</code>	2	when a named function type exists; <code>req</code> , <code>res</code> typed by <code>RequestHandler</code>
<code>app.get("/", async (req, res) => { ... })</code>	callback	no variable; types come from <code>app.get</code>

You will see	Form	Note
<code>ids.map(async (id) => fetchUser(id))</code>	callback	gives <code>Promise<User>[]</code> ; needs <code>Promise.all</code>
<code>const sleep = (ms: number) => new Promise((r) => setTimeout(r, ms))</code>	1	not async, but returns a Promise explicitly